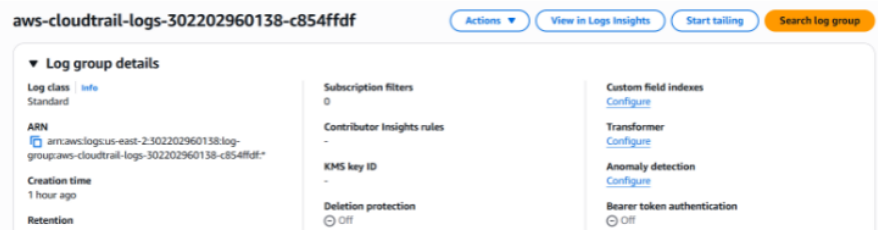
Task on CloudTrail & CloudWatch

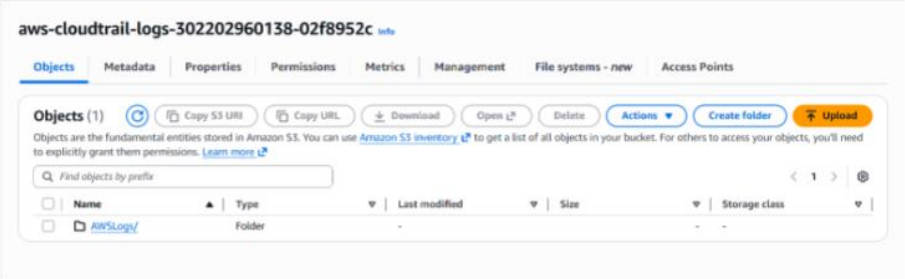
1. Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

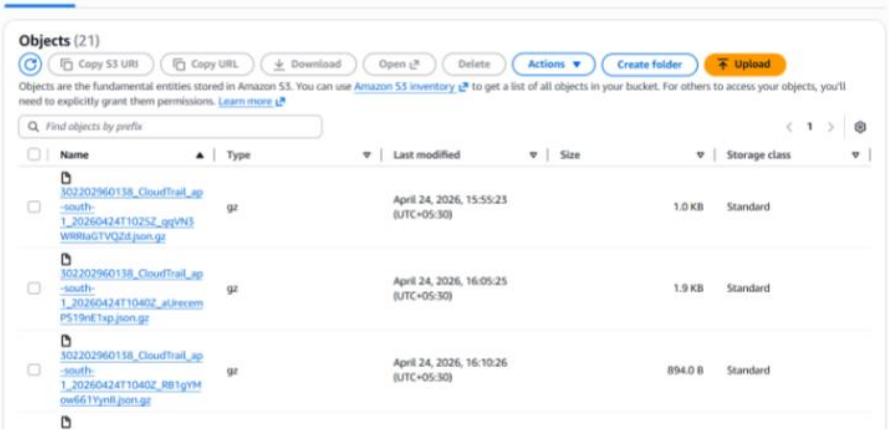
Create cloud trail in that add s3 bucket and cloudwatch log group for store the logs

Create cloudwatch log group



We can logs in S3 bucket

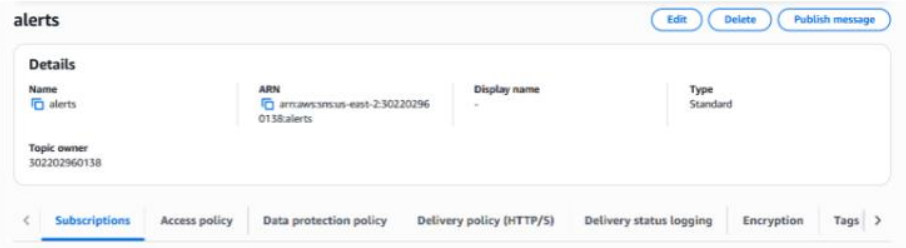




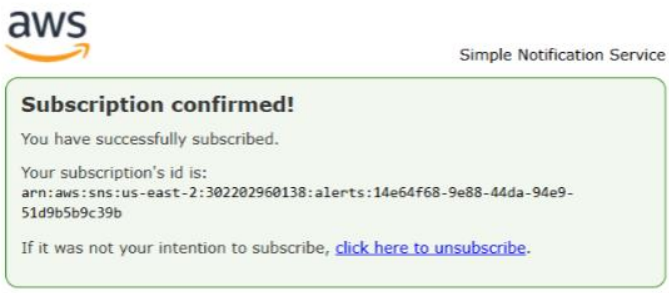
We can check logs in cloud watch log group log steams

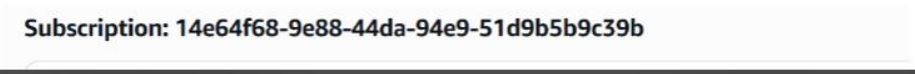
2. Enable SNS for CloudTrail to send alerts via email.

Click to create SNS create a topic
Send the conformation mail to your gmail

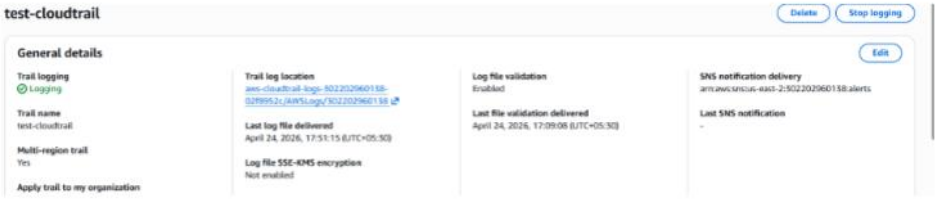


Accept the subscription

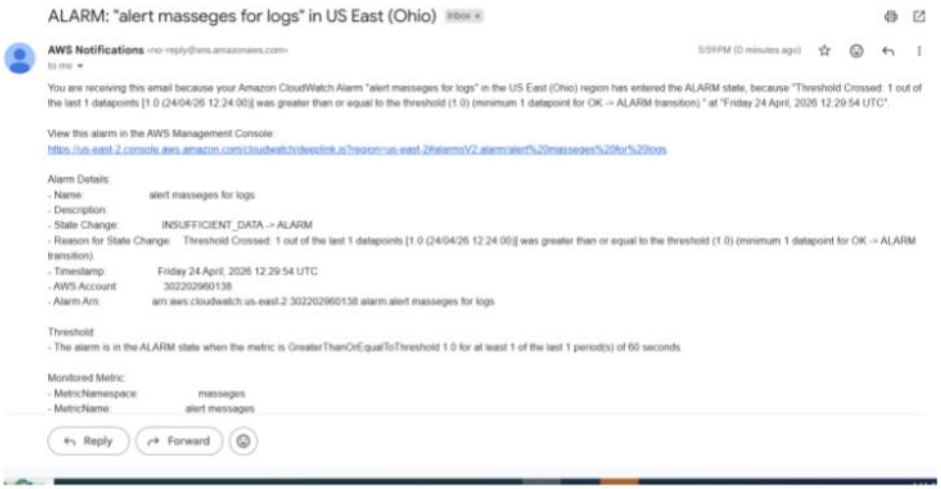




Enable the SNS in the cloud trail



We can see the logs in our mail



Detailed monitoring

After you enable detailed monitoring for an instance, monitoring data is available in 1-minute periods. Learn more

Instance ID

i-0568113d6e9d1e200 (ohio-server)

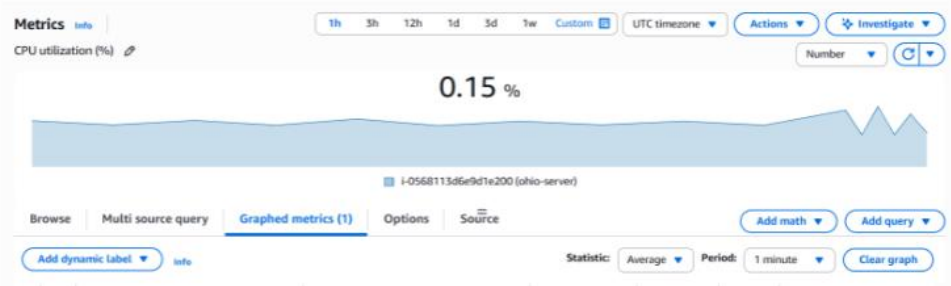
Detailed monitoring

☑ Enable

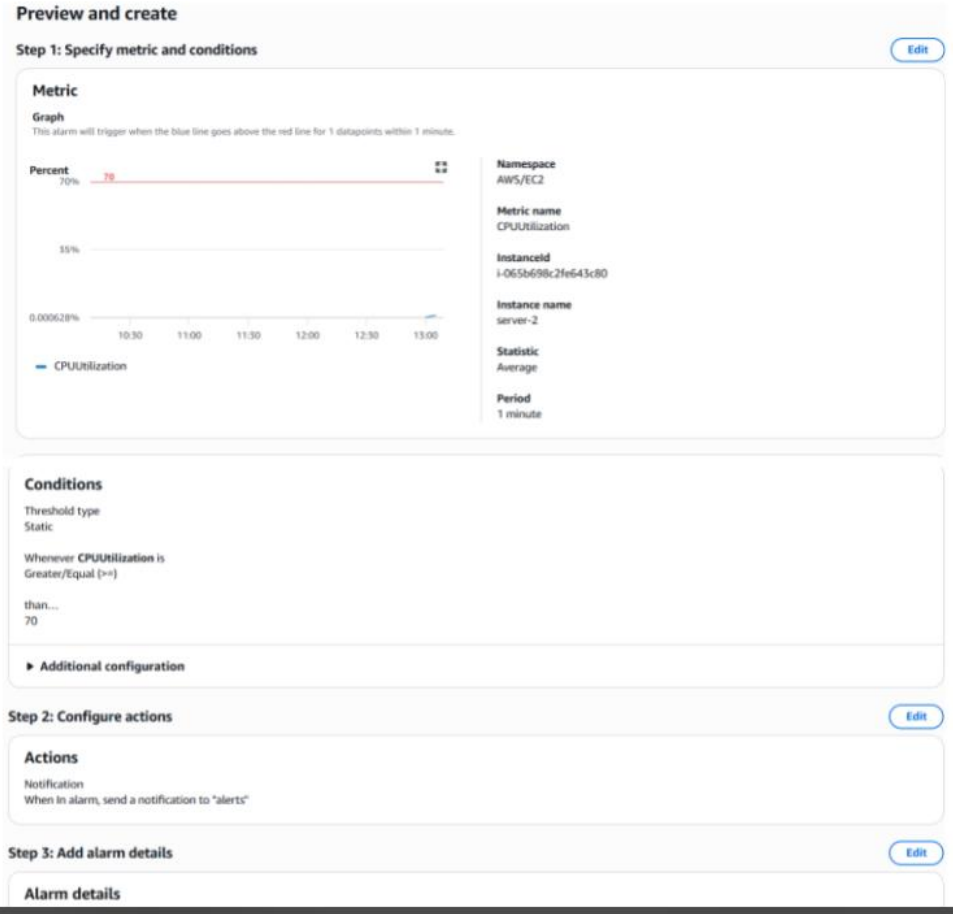
After you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period for the instance.
Additional charges apply

Cancel Confirm

See the metrics of the instance here we can see the detailed monitor



Create alarm — select metric—add metrics



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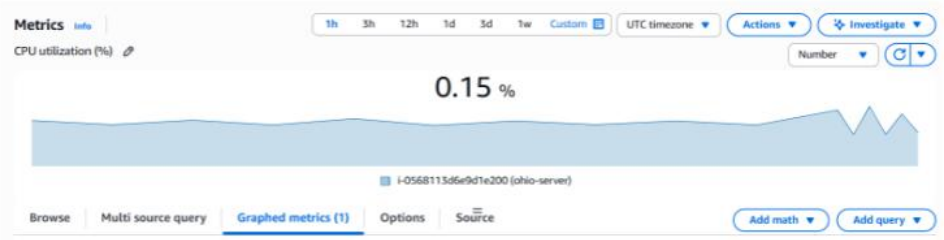

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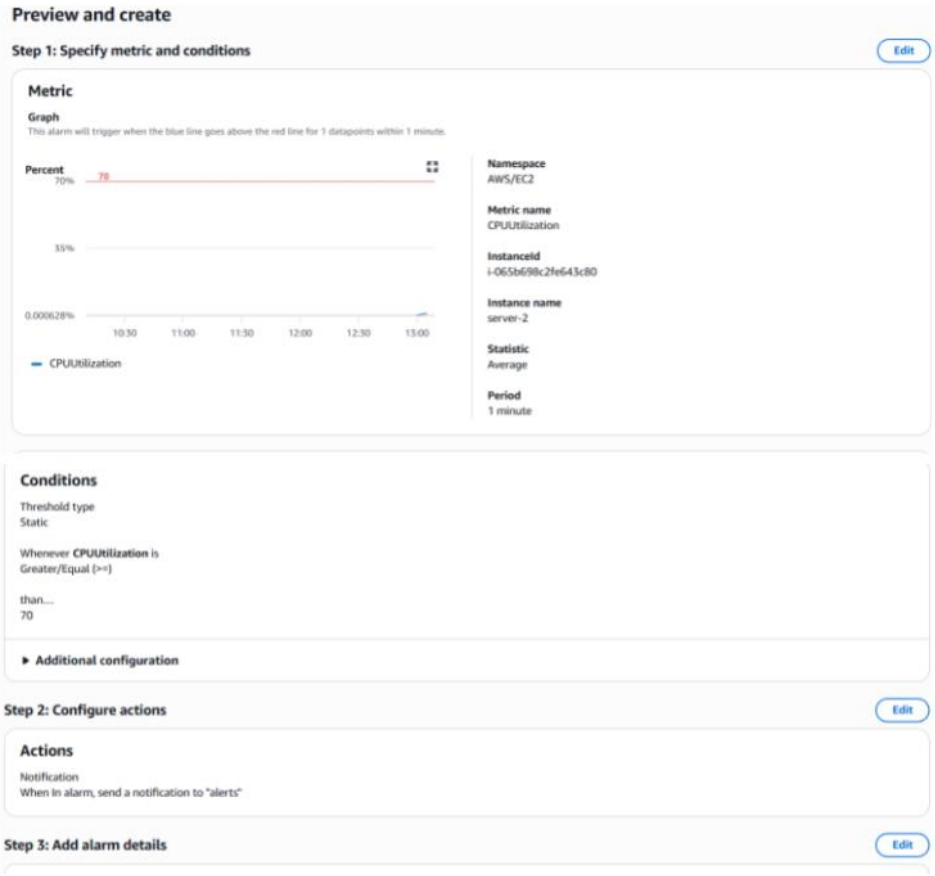
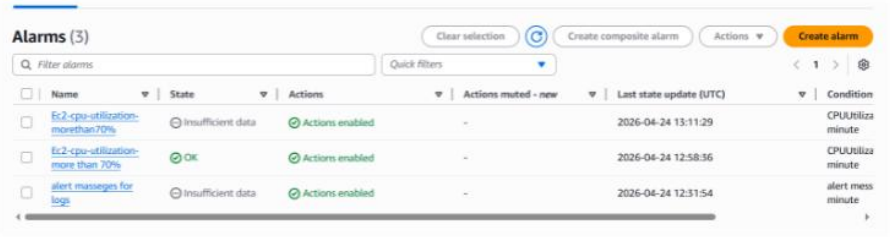




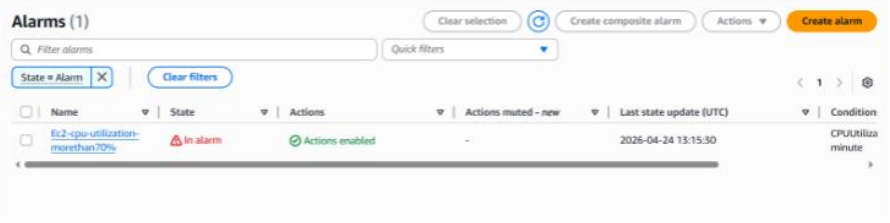
Alarms (3)

Clear selection | Create composite alarm | Actions | Create alarm

Filter alarms | Quick filters

Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
Ec2-cpu-utilization-morethan70%	Insufficient data	Actions enabled	-	2026-04-24 13:11:29	CPUUtiliza minute
Ec2-cpu-utilization-more than 70%	OK	Actions enabled	-	2026-04-24 12:58:36	CPUUtiliza minute
alert masseges for logs	Insufficient data	Actions enabled	-	2026-04-24 12:31:54	alert mess minute

Our EC2 instance CPU utilization more than 70%, now it is alarm stage



Alarms (1)

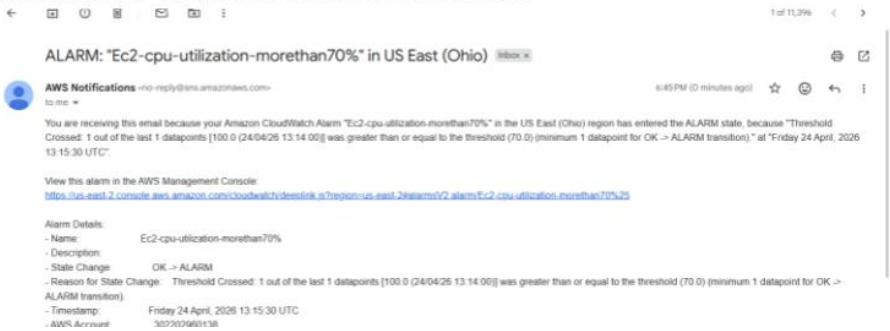
Clear selection | Create composite alarm | Actions | Create alarm

Filter alarms | Quick filters

State = Alarm | Clear filters

Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
Ec2-cpu-utilization-morethan70%	In alarm	Actions enabled	-	2026-04-24 13:15:30	CPUUtiliza minute

Now sent a mail like your CPU utilization more than 70%



ALARM: "Ec2-cpu-utilization-morethan70%" in US East (Ohio) Inbox

AWS Notifications <no-reply@sns.amazonaws.com> 6:45 PM (0 minutes ago)
to me

You are receiving this email because your Amazon CloudWatch Alarm "Ec2-cpu-utilization-morethan70%" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [100.0 (24/04/26 13:14:00)] was greater than or equal to the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Friday 24 April, 2026 13:15:30 UTC".

View this alarm in the AWS Management Console:
https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/Ec2-cpu-utilization-morethan70%25

Alarm Details:
- Name: Ec2-cpu-utilization-morethan70%
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [100.0 (24/04/26 13:14:00)] was greater than or equal to the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition)
- Timestamp: Friday 24 April, 2026 13:15:30 UTC

```
[root@ip-172-31-43-97 scripts]# systemctl status tomcat
○ tomcat.service - Apache Tomcat Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: inactive (dead) since Sat 2026-04-25 09:15:41 UTC; 3min 2s ago
    Duration: 2.267s
    Process: 58670 ExecStart=/root/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
    Process: 58698 ExecStop=/root/tomcat/bin/shutdown.sh (code=exited, status=0/SUCCESS)
   Main PID: 58677 (code=exited, status=0/SUCCESS)
        CPU: 4.763s

Apr 25 09:15:38 ip-172-31-43-97.us-east-2.compute.internal systemd[1]: Starting tomcat.service - Apache Tomcat Web Application Container...
Apr 25 09:15:38 ip-172-31-43-97.us-east-2.compute.internal startup.sh[58670]: Tomcat started.
Apr 25 09:15:38 ip-172-31-43-97.us-east-2.compute.internal systemd[1]: Started tomcat.service - Apache Tomcat Web Application Container.
Apr 25 09:15:41 ip-172-31-43-97.us-east-2.compute.internal shutdown.sh[58704]: NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=jav
Apr 25 09:15:41 ip-172-31-43-97.us-east-2.compute.internal systemd[1]: tomcat.service: Deactivated successfully.
Apr 25 09:15:41 ip-172-31-43-97.us-east-2.compute.internal systemd[1]: tomcat.service: Consumed 4.763s CPU time.
lines 1-15/15 (END)...skipping...
○ tomcat.service - Apache Tomcat Web Application Container
```

Start tomcat service

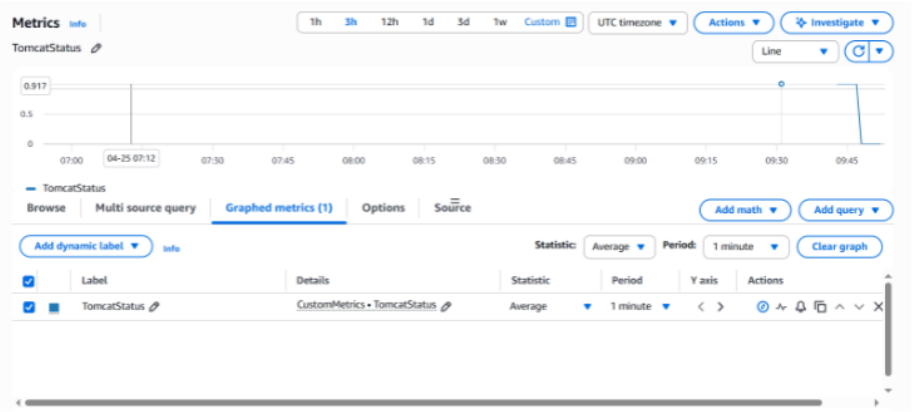
```
[root@ip-172-31-43-97 scripts]# systemctl start tomcat
[root@ip-172-31-43-97 scripts]# systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-04-25 09:19:05 UTC; 4s ago
    Process: 58840 ExecStart=/root/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 58847 (java)
      Tasks: 29 (limit: 1043)
     Memory: 77.7M
        CPU: 4.404s
     CGroup: /system.slice/tomcat.service
             └─58847 /usr/lib/jvm/java-26-amazon-corretto.x86_64/bin/java -Djava.util.logging.config.file=/root/tomcat/conf/logging.propertie

Apr 25 09:19:05 ip-172-31-43-97.us-east-2.compute.internal systemd[1]: Starting tomcat.service - Apache Tomcat Web Application Container...
Apr 25 09:19:05 ip-172-31-43-97.us-east-2.compute.internal startup.sh[58840]: Tomcat started.
Apr 25 09:19:05 ip-172-31-43-97.us-east-2.compute.internal systemd[1]: Started tomcat.service - Apache Tomcat Web Application Container.
```

Create Tomcat Check Script :

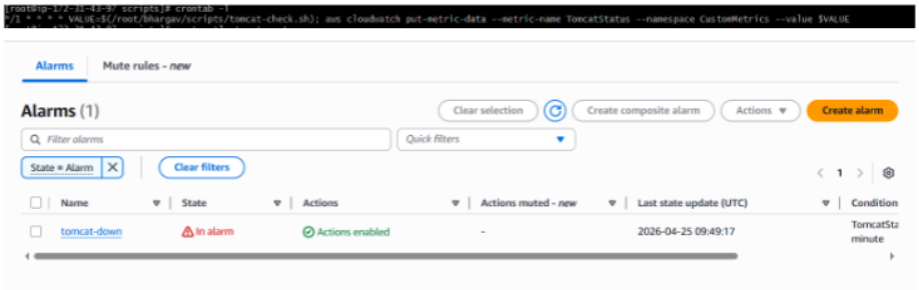
```
[root@ip-172-31-43-97 scripts]# cat tomcat-check.sh
#!/bin/bash
if systemctl is-active --quiet tomcat; then
  echo 1
else
  echo 0
fi
[root@ip-172-31-43-97 scripts]# |
```

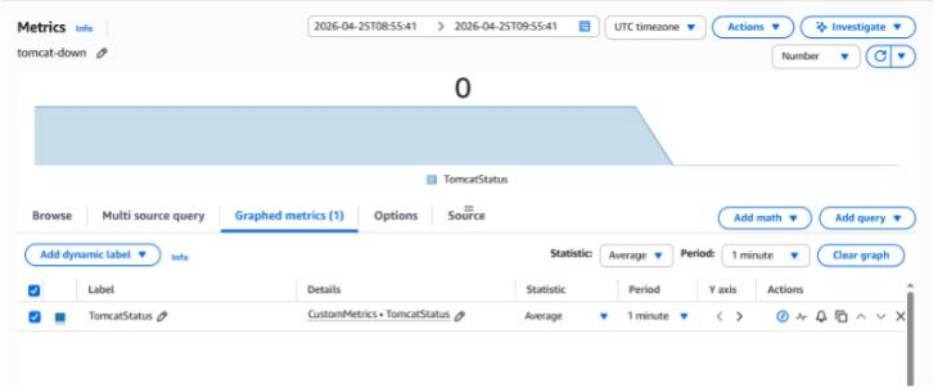
Configure AWS CLI

```
[root@ip-172-31-43-97 ~]# aws configure
AWS Access Key ID [None]: AKIAUMXFTHEFOQ3CREH2
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-2
Default output format [None]: json
```

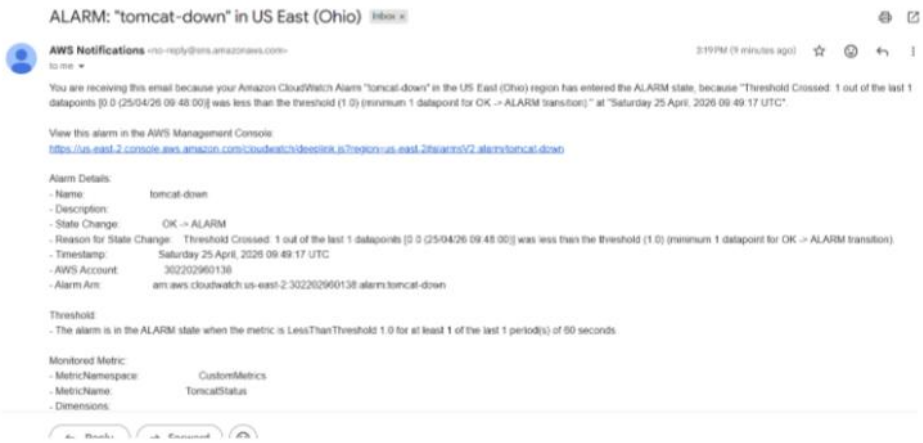


Stop the tomcat service and you get alarm





We received mail like your tomcat service is down



ALARM: "tomcat-down" in US East (Ohio) Inbox x

AWS Notifications <no-reply@sns.amazonaws.com> 3:19 PM (9 minutes ago)
to me

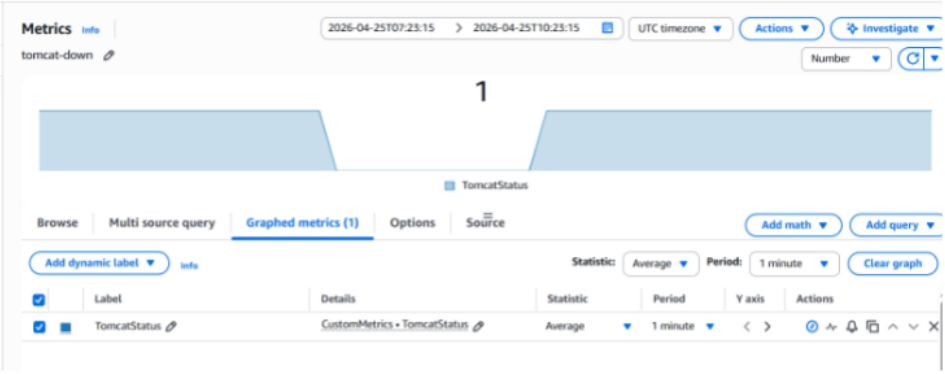
You are receiving this email because your Amazon CloudWatch Alarm "tomcat-down" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [0.0 (25/04/26 09:48:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Saturday 25 April, 2026 09:49:17 UTC".

View this alarm in the AWS Management Console:
https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/tomcat-down

Alarm Details:
- Name: tomcat-down
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [0.0 (25/04/26 09:48:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Saturday 25 April, 2026 09:49:17 UTC
- AWS Account: 302202960138
- Alarm Arn: arn:aws:cloudwatch:us-east-2:302202960138:alarm:tomcat-down

Threshold:
- The alarm is in the ALARM state when the metric is LessThanThreshold 1.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:
- MetricNamespace: CustomMetrics
- MetricName: TomcatStatus
- Dimensions:



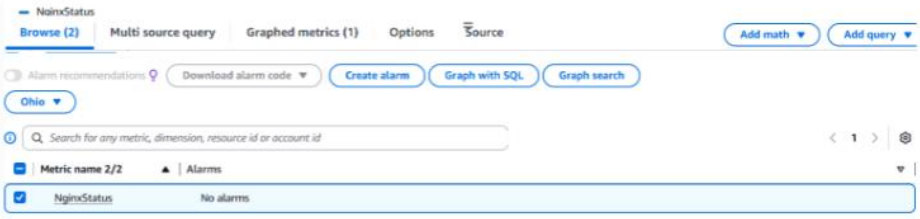
6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

Create Nginx Status Script (on EC2)

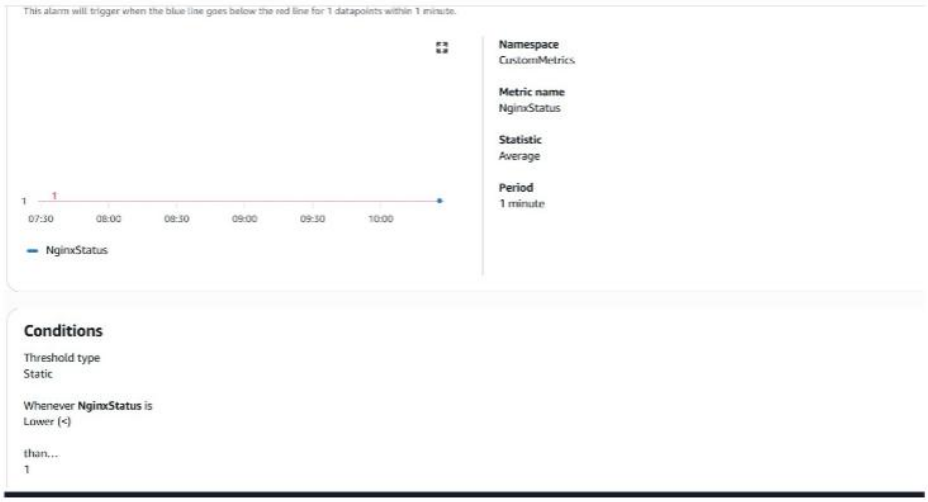
Output:

- 1 → Nginx running
- 0 → Nginx stopped

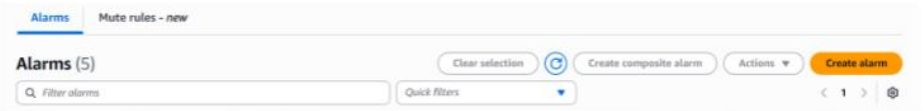
```
[root@ip-172-31-43-97 scripts]# cat nginx-check.sh
ii#!/bin/bash
if systemctl is-active --quiet nginx; then
  echo 1
else
  echo 0
fi
[root@ip-172-31-43-97 scripts]# |
```

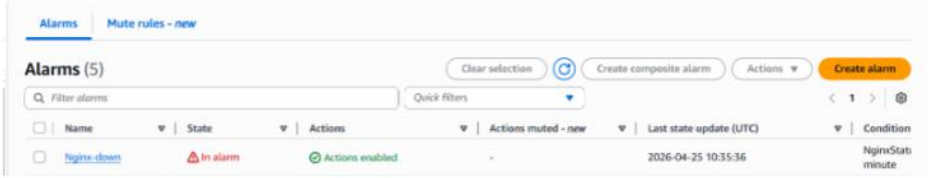


Create alarm



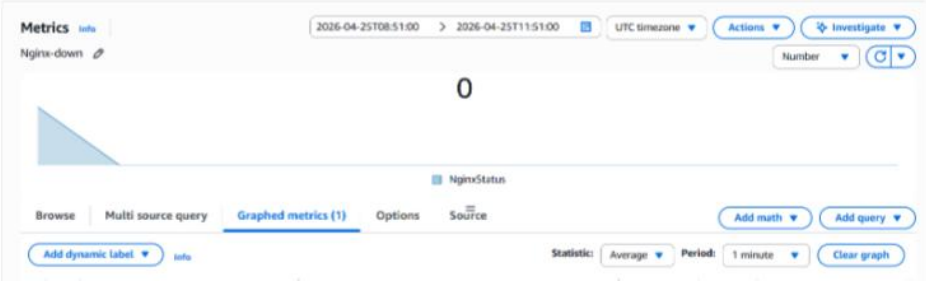
Created an alarm now it is in OK stage

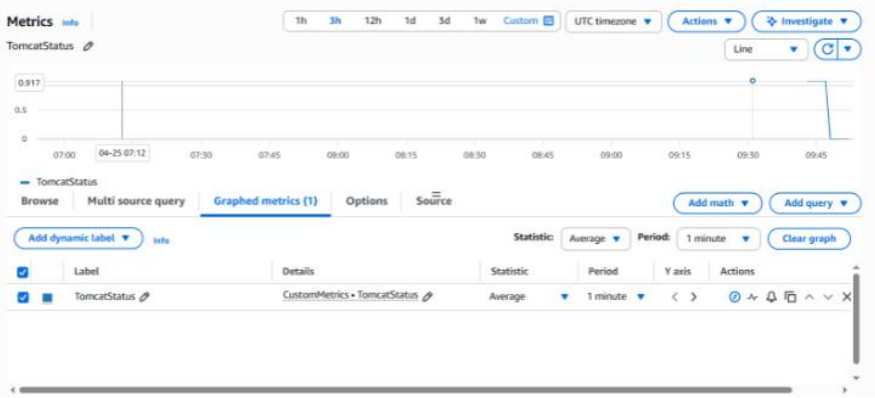
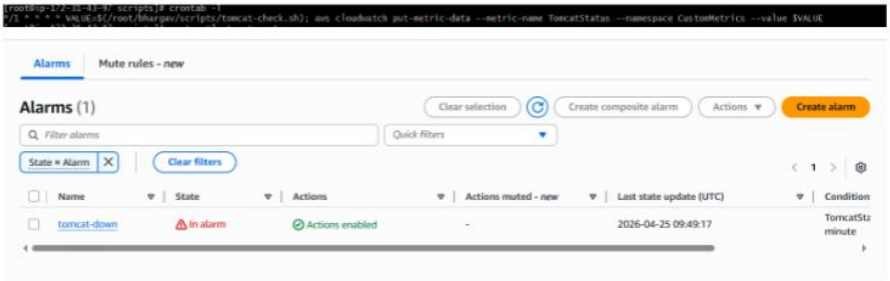




We will the mail

]





Stop the tomcat service and you get alarm